

Logarithms



1

a $\log_5 25 = 2$

c $\log_3 3 = 1$

e $\log_{4.4} 1 = 0$

g $\log_4 32 = \frac{5}{2}$

i $\log_4 \frac{1}{64} = -3$

k $\log_4 9 = m$

m $\log_h w = q$

b $\log_3 81 = x$

d $\log_2 1 = 0$

f $\log_{25} 125 = 1.5$

h $\log_4 0.0625 = -2$

j $\log_x 64 = 1.5$

l $\log_5 q = m$

n $\log_m n = j$

2

a $\log_{10} 100 = 2$

c $\log_{10} (\sqrt{10}) = \frac{1}{2}$

b $\log_{10} \left(\frac{1}{100} \right) = -2$

d $\log_{10} \left(\frac{1}{\sqrt{10}} \right) = -\frac{1}{2}$

3

a $4^2 = 16$

e $3^{-1} = \frac{1}{3}$

i $8^6 = x$

m $r^y = p$

b $5^1 = 5$

f $(5.8)^2 = 33.64$

j $g^3 = 5$

c $8^0 = 1$

g $\left(\frac{1}{3} \right)^{-2} = 9$

k $7^{40} = m$

d $2^{-3} = 0.125$

h $x^5 = 32$

l $y^u = k$

4

a $10^4 = 10\,000$

b $10^{\frac{1}{2}} = \sqrt{10}$

c $10^{-4} = \frac{1}{10\,000}$

d $10^{-\frac{1}{2}} = \frac{1}{\sqrt{10}}$

5 True

Evaluating logarithms

6

a i $x = \log_{10} 820$

ii 2.91

b i $x = \log_{10} 0.0002$

ii -3.70



c i $x = \log_{10} \frac{13}{50}$ ii -0.59

d i $x = \log_{10} 14\,000$ ii 4.15

7

a i 10^6 ii 6 b i 10^{-6} ii -6

c i 10^0 ii 0 d i 10^{-1} ii -1

8

a 6 b -1 c -2 d 0
e 2 f -2 g -2 h $\frac{1}{3}$

i -4

9

a i -1 ii -1 iii -1 iv -1

b 1

10

a 1 b 2 c Between 1 and 2

11

a i between 3 and 4 ii 3.9490

b i between -2 and -1 ii -1.2757

c i between 2 and 3 ii 2.7275

d i between 0 and -1 ii -0.1101

12

a 1.60 b 3.89 c -2.24 d 0.90
e 1.89 f 1.67 g -2.81 h -2.25
i -4.52 j 10.22 k 2.07 l -6.50

13 316.23



14 0.063

15 1